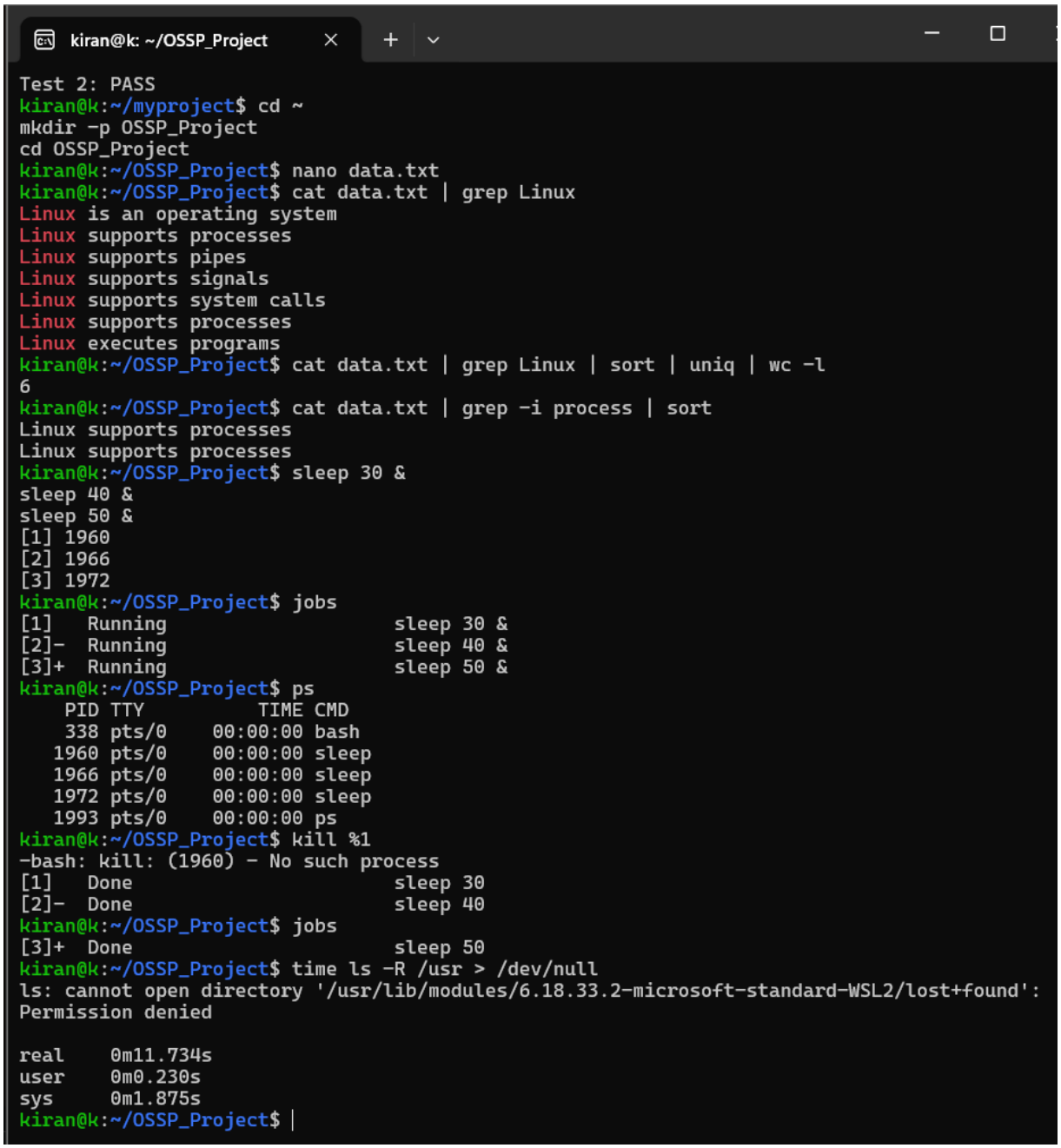
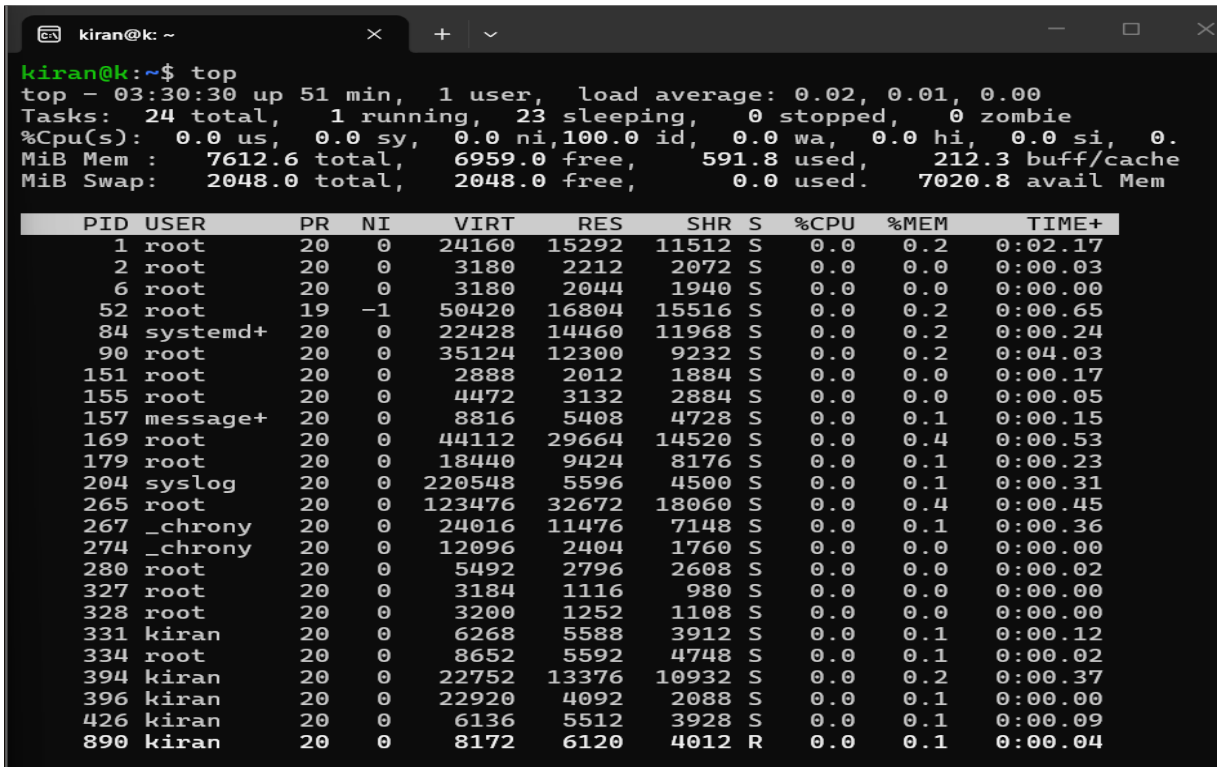
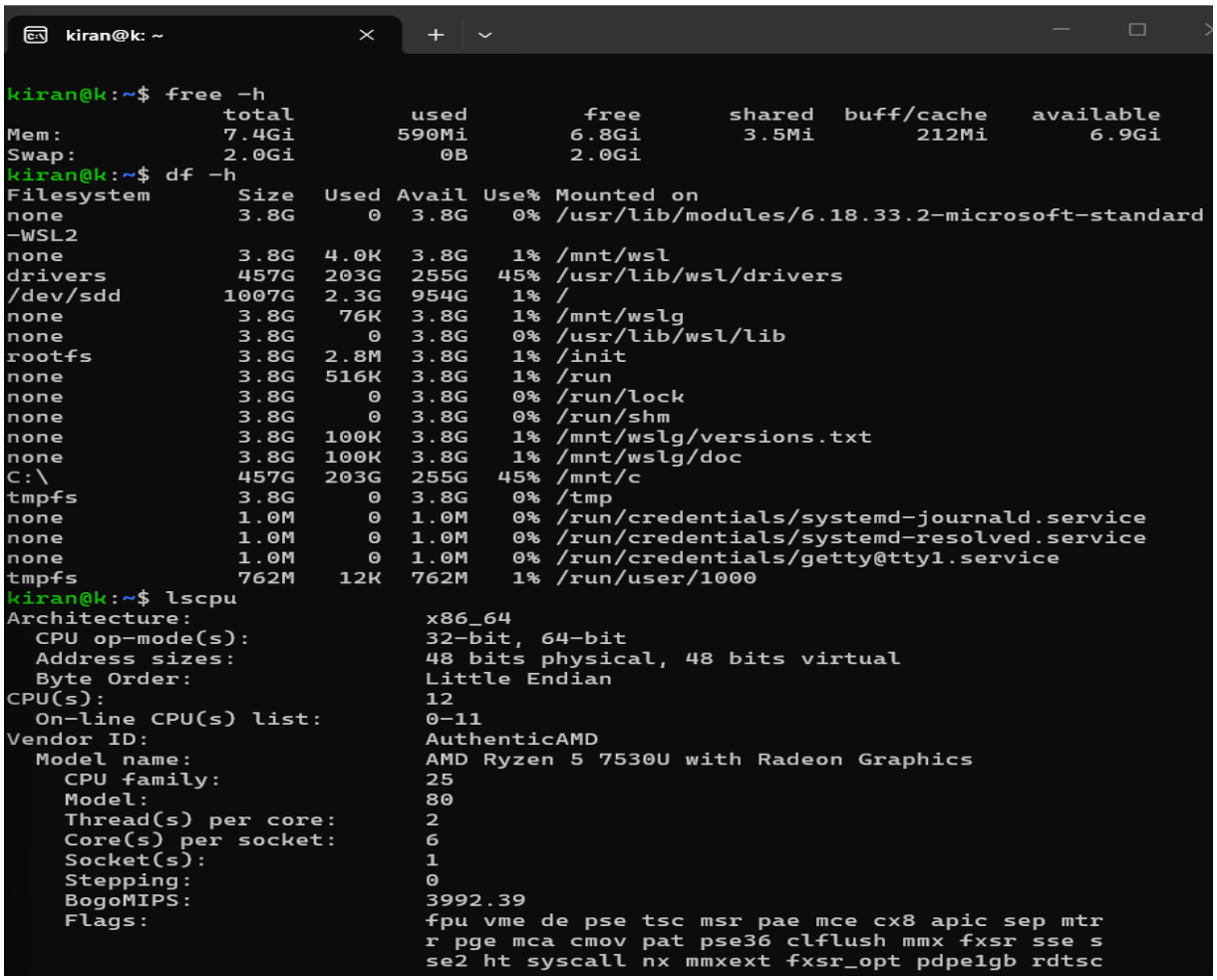
# Skill Week-12

## Session-23

**Task 1**: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.




kiran@k: ~
kiran@k:~$ top
top - 03:30:30 up 51 min,  1 user,  load average: 0.02, 0.01, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :   7612.6 total,   6959.0 free,    591.8 used,    212.3 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7020.8 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+
1 root 20 0 24160 15292 11512 S 0.0 0.2 0:02.17
2 root 20 0 3180 2212 2072 S 0.0 0.0 0:00.03
6 root 20 0 3180 2044 1940 S 0.0 0.0 0:00.00
52 root 19 -1 50420 16804 15516 S 0.0 0.2 0:00.65
84 systemd+ 20 0 22428 14460 11968 S 0.0 0.2 0:00.24
90 root 20 0 35124 12300 9232 S 0.0 0.2 0:04.03
151 root 20 0 2888 2012 1884 S 0.0 0.0 0:00.17
155 root 20 0 4472 3132 2884 S 0.0 0.0 0:00.05
157 message+ 20 0 8816 5408 4728 S 0.0 0.1 0:00.15
169 root 20 0 44112 29664 14520 S 0.0 0.4 0:00.53
179 root 20 0 18440 9424 8176 S 0.0 0.1 0:00.23
204 syslog 20 0 220548 5596 4500 S 0.0 0.1 0:00.31
265 root 20 0 123476 32672 18060 S 0.0 0.4 0:00.45
267 _chrony 20 0 24016 11476 7148 S 0.0 0.1 0:00.36
274 _chrony 20 0 12096 2404 1760 S 0.0 0.0 0:00.00
280 root 20 0 5492 2796 2608 S 0.0 0.0 0:00.02
327 root 20 0 3184 1116 980 S 0.0 0.0 0:00.00
328 root 20 0 3200 1252 1108 S 0.0 0.0 0:00.00
331 kiran 20 0 6268 5588 3912 S 0.0 0.1 0:00.12
334 root 20 0 8652 5592 4748 S 0.0 0.1 0:00.02
394 kiran 20 0 22752 13376 10932 S 0.0 0.2 0:00.37
396 kiran 20 0 22920 4092 2088 S 0.0 0.1 0:00.00
426 kiran 20 0 6136 5512 3928 S 0.0 0.1 0:00.09
890 kiran 20 0 8172 6120 4012 R 0.0 0.1 0:00.04


kiran@k: ~
kiran@k:~$ free -h
total used free shared buff/cache available
Mem: 7.4Gi 590Mi 6.8Gi 3.5Mi 212Mi 6.9Gi
Swap: 2.0Gi 0B 2.0Gi
kiran@k:~$ df -h
Filesystem Size Used Avail Use% Mounted on
none 3.8G 0 3.8G 0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none 3.8G 4.0K 3.8G 1% /mnt/wsl
drivers 457G 203G 255G 45% /usr/lib/wsl/drivers
/dev/sdd 1007G 2.3G 954G 1% /
none 3.8G 76K 3.8G 1% /mnt/wslg
none 3.8G 0 3.8G 0% /usr/lib/wsl/lib
rootfs 3.8G 2.8M 3.8G 1% /init
none 3.8G 516K 3.8G 1% /run
none 3.8G 0 3.8G 0% /run/lock
none 3.8G 0 3.8G 0% /run/shm
none 3.8G 100K 3.8G 1% /mnt/wslg/versions.txt
none 3.8G 100K 3.8G 1% /mnt/wslg/doc
C:\ 457G 203G 255G 45% /mnt/c
tmpfs 3.8G 0 3.8G 0% /tmp
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
none 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.service
none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs 762M 12K 762M 1% /run/user/1000
kiran@k:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 12
On-line CPU(s) list: 0-11
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 5 7530U with Radeon Graphics
CPU family: 25
Model: 80
Thread(s) per core: 2
Core(s) per socket: 6
Socket(s): 1
Stepping: 0
BogoMIPS: 3992.39
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc

```
                          r pge mca cmov pat pse36 clflush mmx fxsr sse s
                          se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc
                          p lm constant_tsc rep_good nopl xtopology tsc_r
                          eliable nonstop_tsc cpuid extd_apicid tsc_known
                          _freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_
                          2 movbe popcnt aes xsave avx f16c rdrand hyperv
                          isor lahf_lm cmp_legacy svm cr8_legacy abm sse4
                          a misalignsse 3dnowprefetch osvw topoext perfct
                          r_core ssbd ibrs ibpb stibp vmmcall fsgsbase bm
                          i1 avx2 smep bmi2 erms invpcid rdseed adx smap
                          clflushopt clwb sha_ni xsaveopt xsavec xgetbv1
                          xsaves clzero xsaveerptr arat npt nrip_save tsc
                          _scale vmcb_clean flushbyasid decodeassists pau
                          sefilter pfthreshold v_vmsave_vmload umip vaes
                          vpclmulqdq rdpid fsrm
Virtualization features:
  Virtualization:         AMD-V
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    128 KiB (4 instances)
  L1i:                    128 KiB (4 instances)
  L2:                     2 MiB (4 instances)
  L3:                     8 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-7
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Not affected
  Spec rstack overflow:   Vulnerable: Safe RET, no microcode
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled v
                          ia prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user
                           pointer sanitization
  Spectre v2:             Mitigation; Retpolines; IBPB conditional; IBRS_
                          FW; STIBP always-on; RSB filling; PBRSB-eIBRS N
                          ot affected; BHI Not affected
  Srbds:                  Not affected
  Tsa:                    Vulnerable: No microcode
  Tsx async abort:        Not affected
```

```
kiran@k:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  24160 15292 ?        Ss   02:39   0:02 /sbin/ini
root           2  0.0  0.0   3180  2212 hvc0     Sl+  02:39   0:00 /init
root           6  0.0  0.0   3180  2044 hvc0     Sl+  02:39   0:00 plan9 --c
root          52  0.0  0.2  50420 16804 ?        S<s  02:39   0:00 /usr/lib/
systemd+      84  0.0  0.1  22428 14460 ?        Ss   02:39   0:00 /usr/lib/
root          90  0.1  0.1  35124 12300 ?        Ss   02:39   0:04 /usr/lib/
root         151  0.0  0.0   2888  2012 ?        Ss   02:39   0:00 /bin/sh /
root         155  0.0  0.0   4472  3132 ?        Ss   02:39   0:00 /usr/sbin
message+     157  0.0  0.0   8816  5408 ?        Ss   02:39   0:00 @dbus-dae
root         169  0.0  0.3  44112 29664 ?        Ss   02:39   0:00 /usr/bin/
root         179  0.0  0.1  18440  9424 ?        Ss   02:39   0:00 /usr/lib/
syslog       204  0.0  0.0 220548  5596 ?        Ssl  02:39   0:00 /usr/sbin
root         265  0.0  0.4 123476 32672 ?        Ssl  02:39   0:00 /usr/bin/
_chrony      267  0.0  0.1  24016 11476 ?        S    02:39   0:00 /usr/sbin
_chrony      274  0.0  0.0  12096  2404 ?        S    02:39   0:00 /usr/sbin
root         280  0.0  0.0   5492  2796 tty1     Ss+  02:39   0:00 /usr/sbin
root         327  0.0  0.0   3184  1116 ?        Ss   02:39   0:00 /init
root         328  0.0  0.0   3200  1252 ?        S    02:39   0:00 /init
kiran        331  0.0  0.0   6268  5588 pts/0    Ss   02:39   0:00 -bash
root         334  0.0  0.0   8652  5592 ?        Ss   02:39   0:00 login --
kiran        394  0.0  0.1  22752 13376 ?        Ss   02:39   0:00 /usr/lib/
kiran        396  0.0  0.0  22920  4092 ?        S    02:39   0:00 (sd-pam)
kiran        426  0.0  0.0   6136  5512 pts/1    Ss+  02:39   0:00 -bash
kiran        932 20.0  0.0   7160  4364 pts/0    R+   03:30   0:00 ps aux
kiran@k:~$
```

```
  GNU nano 8.7.1
Linux is an operating system
Linux supports processes
Linux supports pipes
Linux supports signals
Linux supports system calls
Operating systems manage resources
Processes execute programs
Pipes connect commands
Signals notify processes
```

**Task 2:** To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
kiran@k:~/OSSP_Project/documentation$ nano flow.txt
kiran@k:~/OSSP_Project/documentation$ cat flow.txt
START
  |
  v
Read User Command
  |
  v
Parse the Command
  |
  v
Is there a Pipe?
 /          \
Yes          No
 |            |
 v            v
Create Pipes  Execute Command
      \       /
       \     /
        v   v
     Create Processes
           |
           v
     Execute Commands
           |
           v
       Wait for Jobs
           |
           v
       Display Output
           |
           v
          END
```

```
  GNU nano 8.7.1
# OSSP Project Architecture

## 1. User
The user enters commands through the terminal.

## 2. Shell
The shell receives and parses the commands.

## 3. Command Parser
The parser identifies commands, arguments, pipes and redirections.

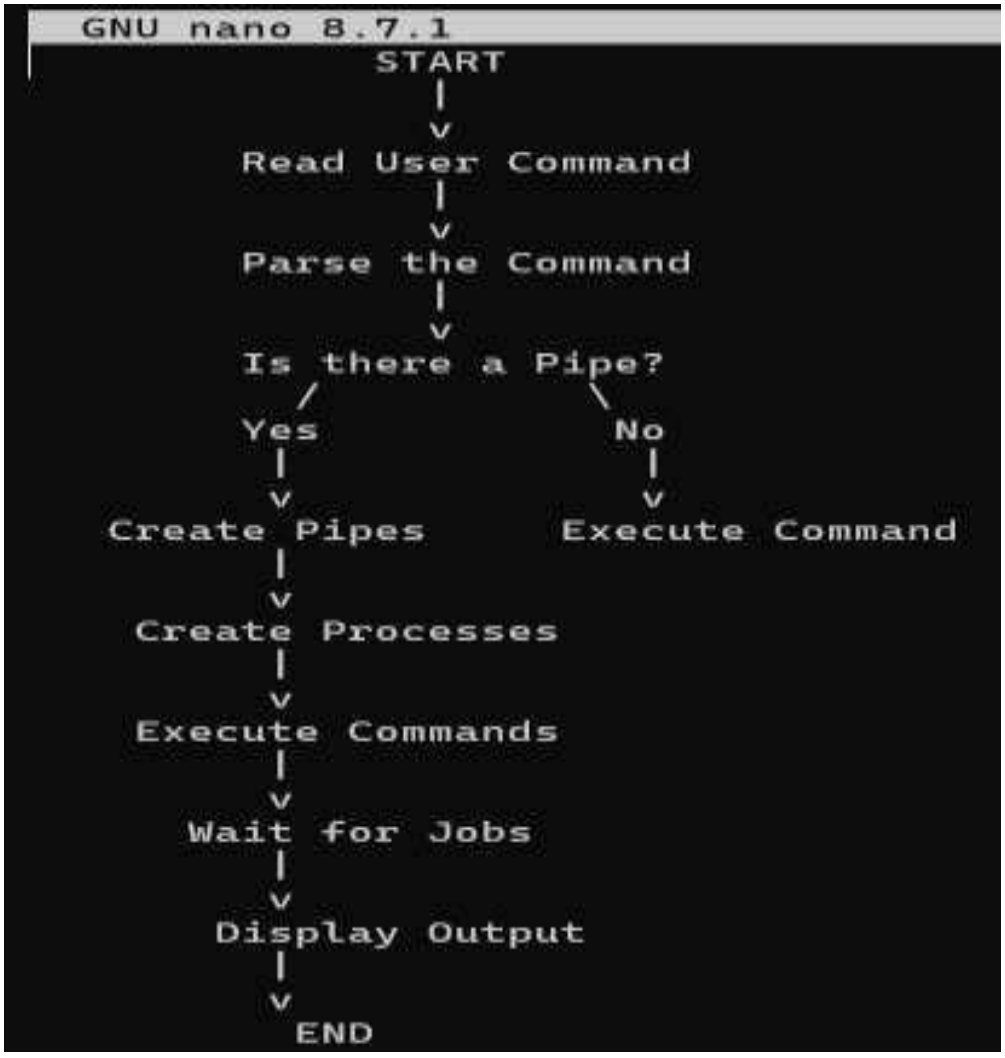
## 4. Process Creation
The shell creates child processes using system calls.

## 5. Execution
Child processes execute the required commands.

## 6. Communication
Pipes allow communication between processes.

## 7. Signals
Signals are used to control or notify processes.

## 8. Output
The final result is displayed on the terminal.
```



```
  GNU nano 8.7.1
# Execution Path

1. User enters a command.
2. Shell reads the command.
3. Shell parses the command.
4. If the command is external, a child process is created.
5. The child process executes the program.
6. Parent process waits for completion.
7. Output is displayed to the user.
8. The shell waits for the next command.
```

```
  GNU nano 8.7.1
# System Calls Used

## fork()
Creates a new child process.

## exec()
Replaces the current process with a new program.

## waitpid()
Allows the parent process to wait for a child process.

## pipe()
Creates a communication channel between processes.

## open()
Opens a file.

## read()
Reads data from a file.

## write()
Writes data to a file.

## close()
Closes an opened file descriptor.

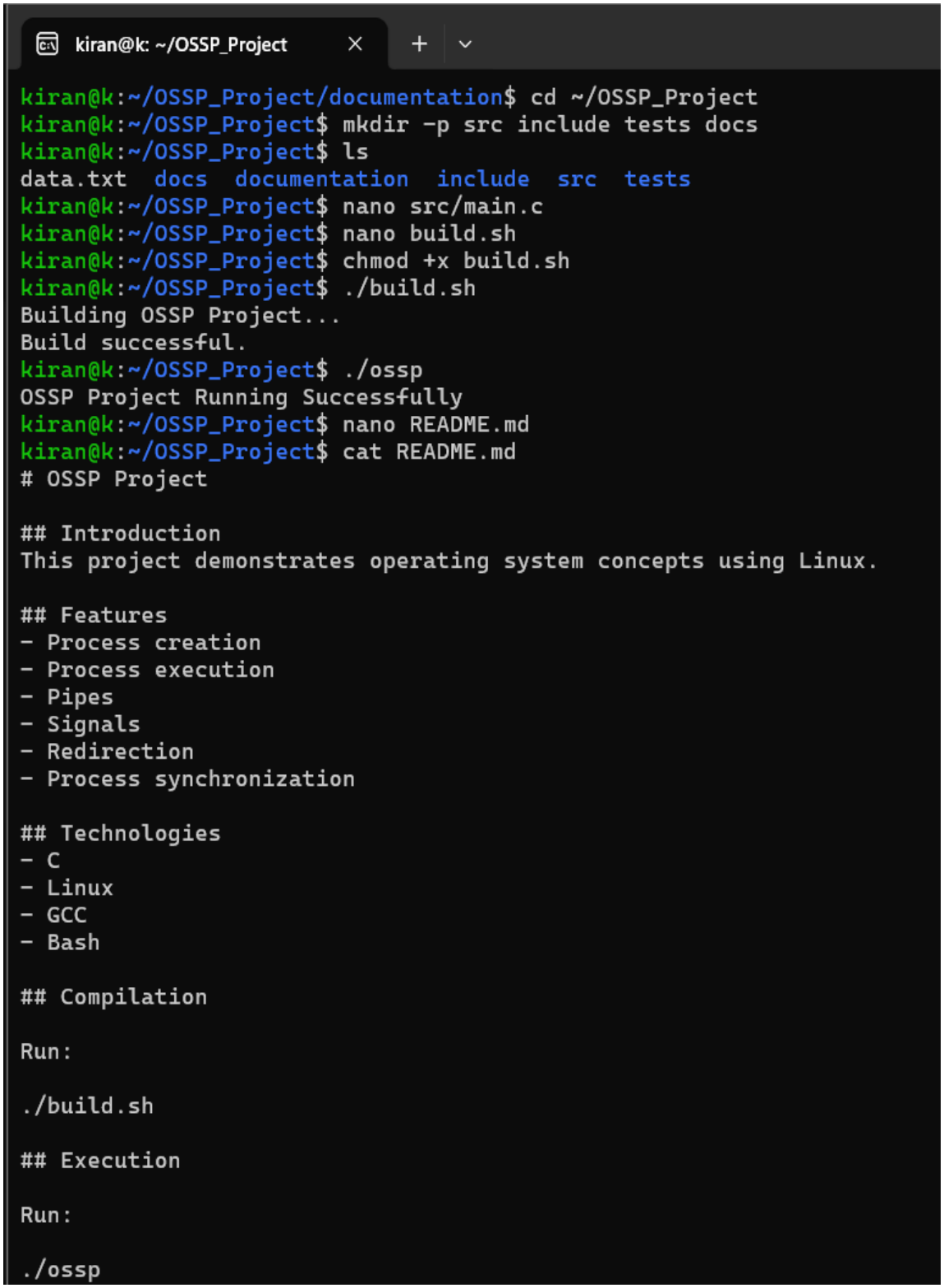
## dup2()
Duplicates a file descriptor and is commonly used for redirection.
```

```
  GNU nano 8.7.1
# Design Decisions

1. Pipes are used for communication between processes.
2. fork() is used to create child processes.
3. exec() is used to execute external commands.
4. waitpid() is used to synchronize parent and child processes.
5. dup2() is used for input and output redirection.
6. Signals are used for process control.
7. Error messages are displayed when commands fail.
```

**Session-24**

**Task 1:** To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.



```
kiran@k:~/OSSP_Project$ find . -maxdepth 2 -type f
./documentation/syscalls.md
./documentation/flow.txt
./documentation/design_decisions.md
./documentation/architecture.md
./documentation/execution_path.md
./build.sh
./data.txt
./README.md
./src/main.c
./ossp
kiran@k:~/OSSP_Project$ ls documentation
architecture.md  design_decisions.md  execution_path.md  flow.txt  syscalls.md
```

```
  GNU nano 8.7.1
#include <stdio.h>

int main()
{
    printf("OSSP Project Running Successfully\n");
    return 0;
}
```

```
  GNU nano 8.7.1
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o ossp

if [ $? -eq 0 ]
then
    echo "Build successful."
else
    echo "Build failed."
fi
```

```
  GNU nano 8.7.1
# OSSP Project

## Introduction
This project demonstrates operating system concepts using Linux.

## Features
- Process creation
- Process execution
- Pipes
- Signals
- Redirection
- Process synchronization

## Technologies
- C
- Linux
- GCC
- Bash

## Compilation

Run:

./build.sh

## Execution

Run:

./ossp
```

**Task2:**To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

